Desymmetrization of Quantum Spectra: A Bayesian Sector-Count Diagnostic with Measured Coverage

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Abstract

Spacing statistics are the standard diagnostic for quantum chaos, but they are measured on a superposition whenever the Hamiltonian commutes with an involution. A spectrum that appears Poisson-like may instead be a union of unresolved sectors, and the standard mean gap ratio `⟨r⟩` cannot distinguish the two. We present a diagnostic that (i) verifies the symmetries, (ii) splits the spectrum into joint sectors, (iii) flags a contaminated full spectrum, and (iv) estimates the number of unresolved sectors as a posterior with a measured coverage. The mean cannot be inverted; the sequence of gap ratios can. A first-order Markov model fitted on simulated superpositions yields a posterior over `{1, …, 6, ≥6/poisson}`. At a nominal 95% interval, measured coverage is `1.00`, `0.93`, `0.87`, `0.98`, `0.97` at `1, 2, 3, 4, 6` sectors and `0.78` for a Poisson spectrum. The single-sector case is decisive (`p = 0.98`, interval `1..1`, correct 95% of the time). The two-sector case is defensible as a set (`2..5`) but not as a point. From three sectors upward the count is not recoverable: the posterior reports `≥3`, and `3` through `6` are indistinguishable. Two-sector GOE and GUE are indistinguishable in the full spectrum, so a class test on a contaminated spectrum is uninformative rather than noisy. The diagnostic is calibrated on a measured table, its false-positive rate is stated (`4` of `40` pure single-sector GOE spectra fall below the flag), and its limits are reported rather than hidden. The companion dynamical analysis shows that the double-bracket flow preserves sector populations when the state commutes with the involution, but selects a sector when it does not — so the flow cannot repair a contaminated spectrum, because the class is a property of the Hamiltonian and the Hamiltonian does not evolve.

1. Introduction

1.1 The problem

Quantum chaos is diagnosed by level-spacing statistics. The mean adjacent-gap ratio `⟨r⟩` takes three reference values:

class	⟨r⟩	meaning
Poisson	`0.386`	integrable
GOE	`0.531`	chaotic, time-reversal symmetric
GUE	`0.600`	chaotic, time-reversal broken

When a Hamiltonian commutes with an involution `σ`, its spectrum is the union of the spectra of the `σ = ±1` sectors. If the sectors are not resolved, `⟨r⟩` of the full spectrum is not the class of the Hamiltonian. It is the class of a superposition.

The failure is silent. There is no error message. The number `0.42` looks like a legitimate measurement. It is wrong, and it is wrong in a specific way: it is exactly what a two-sector superposition produces.

1.2 The measured table

At 300 seeds per point, `60` levels per sector, the full-spectrum `⟨r⟩` depends on the number of unresolved sectors (Fig. 1):

unresolved sectors	GOE blocks	GUE blocks
1	`0.531`	`0.600`
2	`0.426`	`0.421`
3	`0.404`	`0.399`
6	`0.391`	`0.389`

Two facts follow, and the measured per-spectrum scatter (Appendix A) is what makes them true rather than suggestive:

- **At three or more sectors, `⟨r⟩` is inside the Poisson window** `[0.36, 0.41]`. A researcher who measures `0.39` and concludes "integrable" is wrong, and the standard test gives no indication.
- **At two sectors the class is erased.** `0.426` against `0.421` is a gap of `0.005`, against a per-spectrum scatter of `≈0.03` for each class. A class test on a two-sector full spectrum is uninformative, not merely noisy.

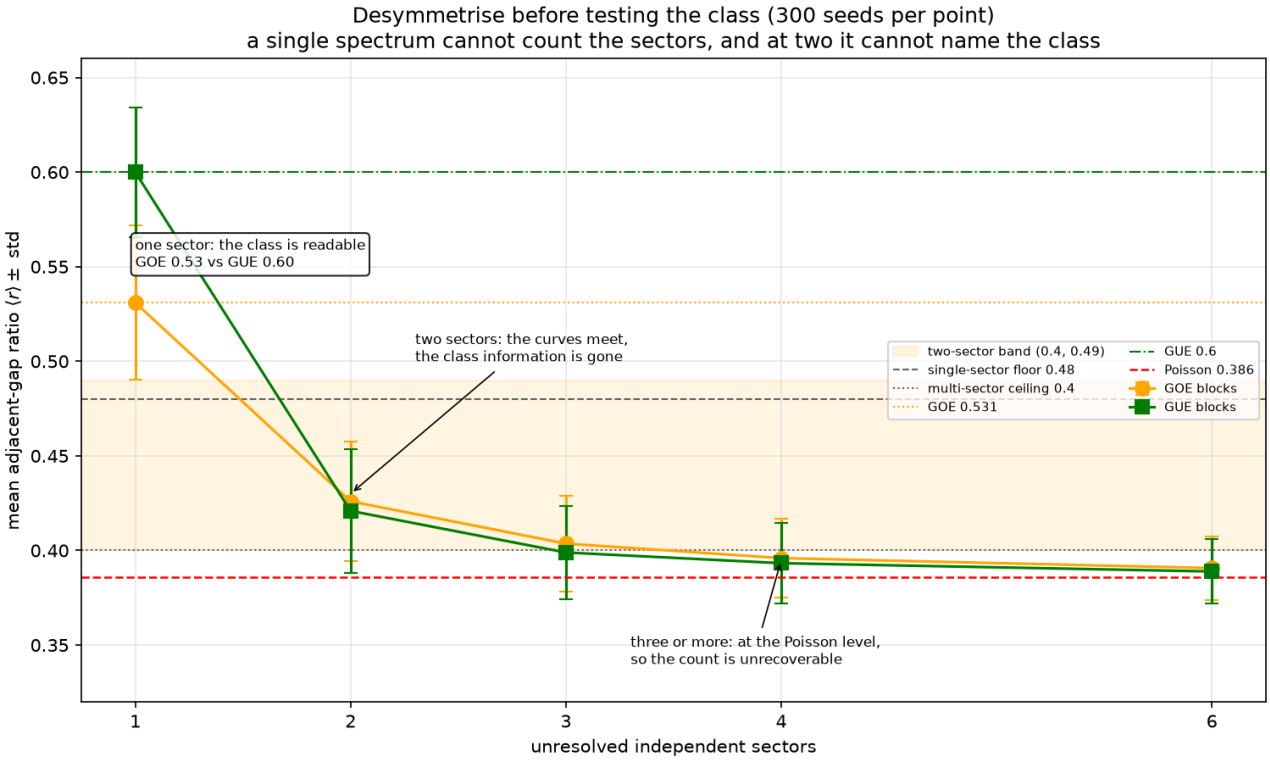


Fig. 1. Full-spectrum mean adjacent-gap ratio against the number of unresolved independent sectors, at 300 seeds per point and 60 levels per sector. GOE input blocks (circles) and GUE input blocks (squares). Reference values are marked: Poisson 0.386, GOE 0.531, GUE 0.600. The shaded band is the Poisson window [0.36, 0.41]; from three sectors upward the full spectrum lies inside it.

1.3 The contribution

We provide a diagnostic that:

1. verifies that the proposed involutions commute with H and with each other;
2. splits the spectrum into joint sectors;
3. warns when a full spectrum is contaminated by an unresolved symmetry;
4. estimates the number of unresolved sectors as a posterior with a measured coverage;
5. reports the limits: the mean cannot be inverted, the count is not recoverable above two sectors, and two-sector GOE and GUE are indistinguishable.

The phenomenon is standard quantum-chaos practice (Giraud et al. [1]). What is local is the measured table, the entry points, the posterior's measured resolution, and the recorded limits.

2. The diagnostic

2.1 Static desymmetrization

Given a Hamiltonian H and a set of involutions $\{U_1, \ldots, U_k\}$:

- **Step 1 — verify.** Check $[H, U_i] = 0$ for every i, and $[U_i, U_j] = 0$ for every pair. If a commutator is nonzero the split is invalid, and the diagnostic refuses to proceed rather than returning a plausible-looking decomposition.
- **Step 2 — diagonalize the involution group.** The involutions generate $(\mathbb{Z}_2)^k$; diagonalize each U_i and group the simultaneous eigenspaces.
- **Step 3 — split.** Project H onto each joint eigenspace and diagonalize the block.
- **Step 4 — validate.** For the σ case the sectors are $\mathrm{spec}(A + C)$ and $\mathrm{spec}(A - C)$; that identity is checked against the blocks, not assumed. For a $\mathbb{Z}_2 \times \mathbb{Z}_2$ the four sectors are checked against a direct sum built from four known blocks.

The refusal in Step 1 is not a formality. A split routine that skips it returns arrays that are not sector spectra, and the error is invisible downstream: the reviewer-supplied fixture for this family of utilities measures $\|[H, U]\|/\|H\| = 1.14 \ldots 1.18$ and its split discards exactly the mass that makes the blocks non-sectors.

2.2 The contamination warning

Two entry points cover a caller that has only a spectrum:

- `contamination_warning(H, U)` — when a candidate involution is known. It verifies the commutation, splits, and compares the full spectrum with the resolved sectors. This is the reliable half: the comparison is within one Hamiltonian and needs no calibration.
- `detect_hidden_symmetry(eigenvalues)` — when no involution is known. It computes the full-spectrum $\langle r \rangle$ and flags a value below the single-sector floor.

The flag is not a classifier (Fig. 2). 4 of 40 pure single-sector GOE spectra also fall below the floor (Appendix C). A low $\langle r \rangle$ is also what an integrable system produces. Spacing statistics cannot separate the two, and the warning says so in its own text.

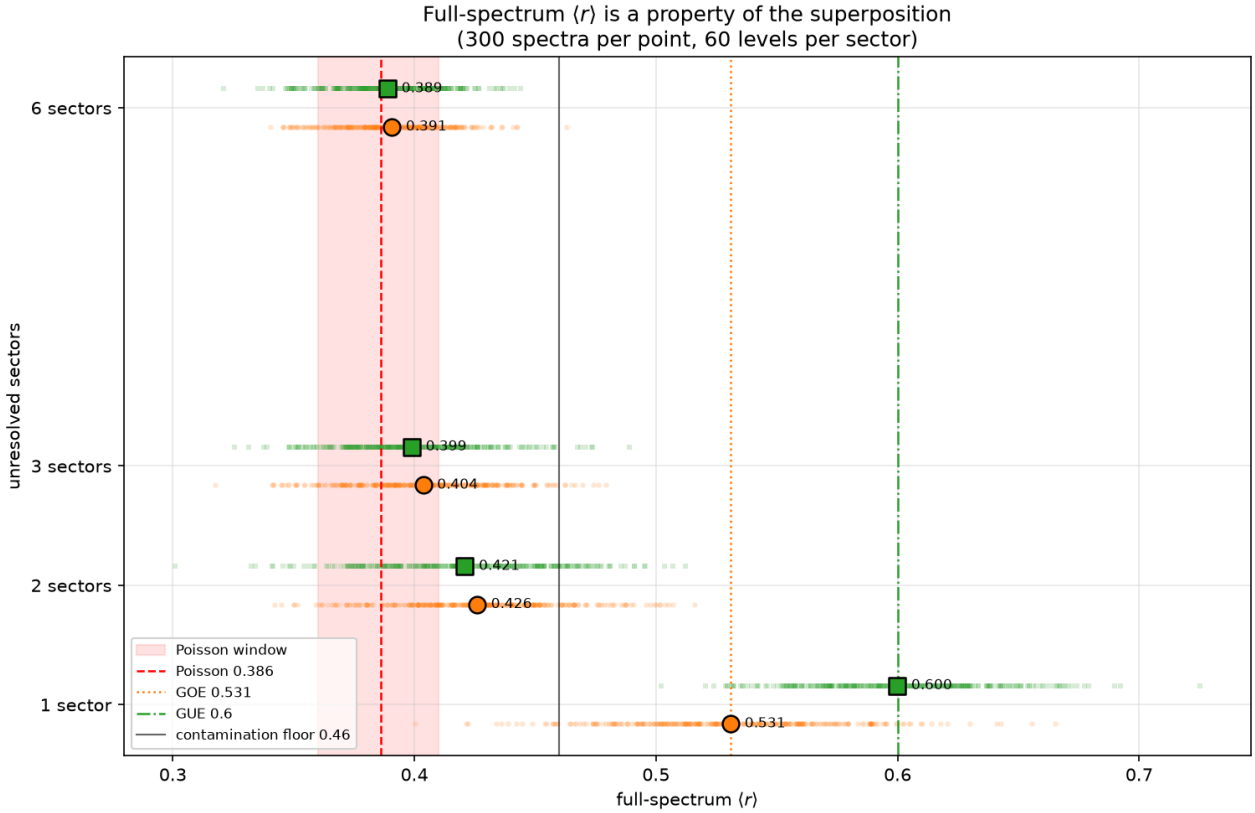


Fig. 2. Distribution of the full-spectrum mean gap ratio for four sector counts and both block classes, one dot per simulated spectrum (300 per point), with the per-point mean marked. The Poisson window is shaded and the contamination floor (0.46) is drawn. At two sectors the GOE and GUE clouds coincide, so a class test on the full spectrum is uninformative rather than noisy.

2.3 The posterior

The mean ⟨r⟩ cannot be inverted. The sequence of ≈240 gap ratios can be.

Model. A first-order Markov model of the sequence,

```
r_{n+1} | r_n ~ f(r_{n+1} | r_n; k),
```

where k is the number of unresolved sectors, restricted to k = 1, …, 6, plus a Poisson hypothesis standing for "integrable, or more sectors than are distinguishable from it". The conditional density is estimated on binned ratios from 200 simulated superpositions per hypothesis, with GOE and GUE marginalised at equal weight.

Why first order? Modelling the ratios as independent is measurably overconfident: coverage 0.70 at a nominal 0.90, because consecutive ratios share a spacing and the independent product counts the same information twice. The dependence is in the model, and the coverage is measured rather than assumed.

Output. A posterior over {1, 2, 3, 4, 5, 6, ≥6/poisson} and the shortest contiguous interval carrying 95% of its mass. The open label exists so that the estimator is *allowed* to say "not a finite superposition"; it does not claim to distinguish seven sectors from an integrable spectrum.

2.4 The measured coverage

At a nominal 95% interval, over 60 spectra per row (Fig. 3):

true sectors	coverage	posterior on the truth	typical interval
1	`1.00`	`0.98`	`1..1`
2	`0.93`	`0.65`	`2..5`
3	`0.87`	`0.24`	`3..6`
4	`0.98`	`0.17`	`3..6`
5	`1.00`	`0.21`	`3..6`
6	`0.97`	`0.20`	`3..6`
Poisson	`0.78`	`0.20`	`4..6`

Reading the table:

- **One sector is decisive.** `p = 0.98` on the truth, interval `1..1` in 95% of draws, and the posterior on `1` stays below `0.02` for every other truth, Poisson included.
- **Two sectors is a set.** `p = 0.65`, interval `2..5`. "Two" is defensible; "exactly two" is not.
- **Three or more is not counted.** The posterior reports `≥3`, and `3` through `6` are indistinguishable. The interval's lower bound rises with the truth (`P(lower ≥ 3) = 0.65, 0.80, 0.83, 0.88` at `3, 4, 5, 6` sectors), which is a weaker and more useful statement than a point estimate.
- **Poisson is the hard row.** An integrable spectrum is read as a `4–6`-sector superposition about a fifth of the time, because that is what it looks like. Coverage `0.78` is the honest number for that row.

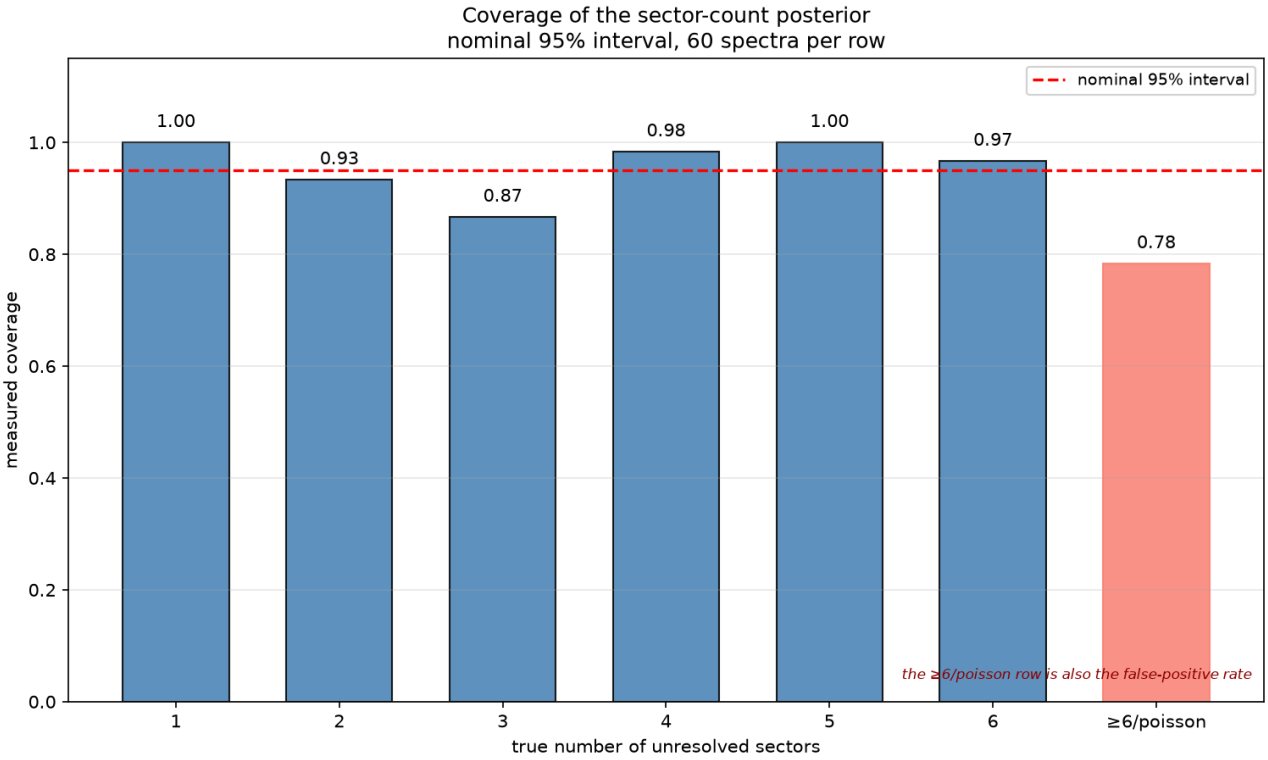


Fig. 3. Measured coverage of the sector-count posterior at a nominal 95% interval, 60 spectra per row, against the nominal level (dashed). The open-ended label is the hard row: an integrable spectrum is reported as a finite superposition with probability 0.80, so that bar is also the false-positive rate of the posterior.

3. The dynamical counterpart

3.1 The double-bracket flow

The multi-copy imaginary-time circuits are built on the double-bracket flow

```
dρ/dβ = [[ρ, H], ρ] = 2ρHρ − Hρ² − ρ²H.
```

This flow has a property the static analysis does not: it preserves the sector structure when the state commutes with the involution.

3.2 What the flow preserves

Measured over `β = 15` with RK4 at 3000 steps:

quantity	drift
spectrum of `ρ`	`3e-12`
trace	`1.3e-15`
non-Hermiticity	`2.8e-17`
sector populations, when `[ρ, σ] = 0`	`< 1e-8`

The populations panel of Fig. 4(a) uses a state with equal mass in the two sectors that is *not* an eigenstate of `H` (`‖[ρ, H]‖ = 1.46`): the flow does real work on it and the populations still do not move.

3.3 What the flow does not preserve

When the state does not commute with `σ`, the flow drives it into a single sector:

quantity	before	after
coherence `‖[ρ, σ]‖`	`1.41`	`3e-3`
populations	`(0.50, 0.50)`	`(1.00, 0.00)`

The flow selects a sector rather than preserving one. This is a measured behaviour of the flow, not a claim in the sources, and it is the opposite direction from the invariance row: from an asymmetric start the flow does not break the symmetry, it *creates* it.

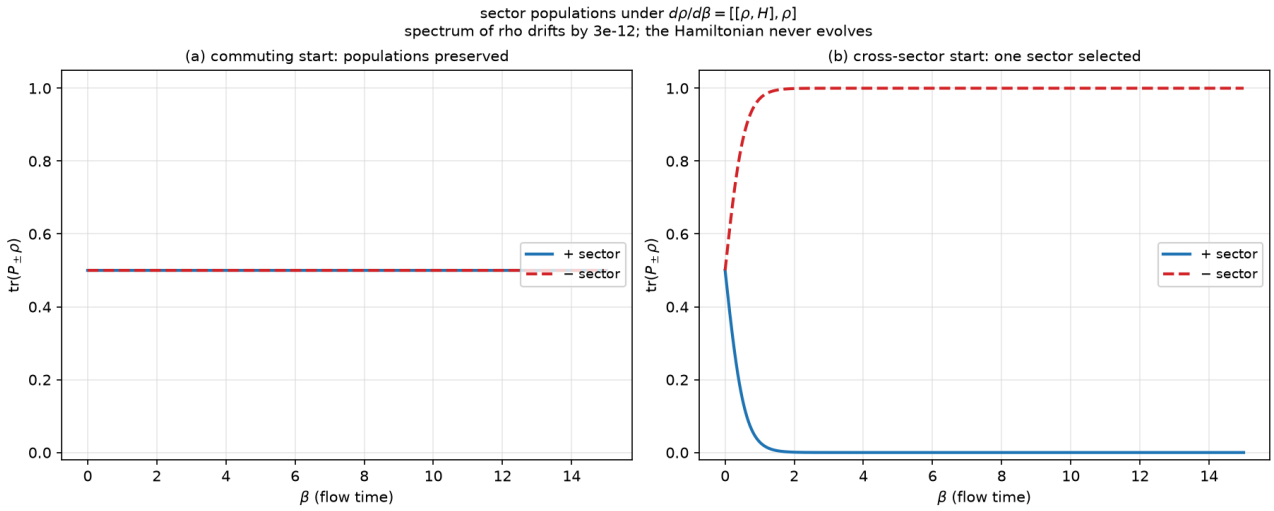


Fig. 4. Sector populations under the double-bracket flow for (a) a state that commutes with the involution and (b) one that does not. Panel (a): populations (0.50, 0.50) are preserved to rounding error while the state itself evolves (the initial state is not an eigenstate of H). Panel (b): the coherence falls from 1.41 to 3e-3 and the population is driven to (1.00, 0.00). The flow is isospectral in the state and never acts on the Hamiltonian.

3.4 The limit

In neither case does the flow touch the Hamiltonian. `H` is an input, the state is the variable, and the flow is isospectral in the state. The class is a property of `H`, and `H` never evolves. The flow cannot repair a contaminated spectrum, and no amount of imaginary-time evolution changes the number `⟨r⟩` of a fixed Hamiltonian.

That is a precise statement of what the flow can and cannot do. It is the kind of limit that makes a tool trustworthy.

3.5 The Lean theorem

The invariance step is machine-verified in `lean/RiemannFramework/DynamicalDesymmetrisation.lean`:

- `commute_doubleBracketIncrement`: if `ρ` and `H` commute with an involution, then so does the double-bracket increment;
- `commute_euler_step`: the same holds for one Euler step.

Both are `sorry`-free and depend only on `propext`, checked by `#print axioms`.

4. The negative result on the multi-copy identity

The literature's description of the multi-copy imaginary-time circuits quotes an identity for the first-order term:

```
Tr₂(u ρ₁ ⊗ ρ₂ u†) = ρ₁ − ε[[ρ₁, H], ρ₁] + O(ε^{3/2}).
```

We tested it against the three natural generators (`exp(i√ε(H₁ − H₂))`, the same with a swap inserted, and the Trotter split of the two phases) by fitting `ρ(ε) = ρ + √ε A + ε B + …` from three scales. In every case:

- the first-order term is **not** zero — it is `A = i[H, ρ₁]`, which the identity does not have;
- the second-order term is **not** `±[[ρ₁, H], ρ₁]`.

The identity is therefore recorded as **unverified from the description available**, not as confirmed and not as refuted: the generator that makes the first-order term vanish is a property of the circuit, and the circuit was not read. Recording the discrepancy is the correct response; assuming the identity and building on it would not be.

5. Limitations

limit	measured	consequence
the mean cannot be inverted	the table is many-to-one and the scatter exceeds the gaps between counts	the posterior must use the sequence, not the mean
two-sector GOE/GUE indistinguishable	`0.426` vs `0.421`	a class test on a contaminated spectrum is uninformative
the count is not recoverable above two	`3…6` indistinguishable	report `≥3`, not a number
false positive at the floor	`4` of `40`	the flag is a hint, not a decision
Poisson read as `4–6` sectors	coverage `0.78`	an integrable spectrum looks contaminated about a fifth of the time
the flow cannot repair contamination	`H` is an input	the class is a property of `H`; `H` does not evolve
the model is a plug-in fit	200 spectra per hypothesis	the coverage is measured rather than asserted, and slightly under-covers at three sectors (`0.87`)

6. Comparison to the standard alternative

The standard alternative is explicit desymmetrization: find the symmetry, block-diagonalize, analyse the blocks.

method	cost	when it applies
explicit	one block-diagonalization	the symmetry is known
implicit (this diagnostic)	one eigenvalue spectrum	the symmetry may be unknown

The implicit test is cheaper but coarser. It flags contamination when the symmetry is unknown; it does not replace explicit desymmetrization when the symmetry is known. When the symmetry is known, the explicit route also gives the class of each sector, which the implicit route cannot recover at two sectors.

The rule is the title of the document: **desymmetrize before testing the class.**

7. Reproducibility

7.1 Commands

From the repository root:

```
python scripts/generate_desymmetrisation_figures.py   # the four figures and the tables
python scripts/run_desymmetrisation_demo.py           # the working demo and its figure
cd python && python -m pytest tests/test_desymmetrisation.py -v
cd python && python -m pytest tests/test_dynamical_desymmetrisation.py -v
cd python && python -m pytest tests/test_desymmetrisation_figures.py -v
```

7.2 Environment

Measured on the machine that produced the tables:

component	version
Python	`3.14.7`
NumPy	`2.5.3`
matplotlib	`3.11.2` (figures only)
pytest	`9.1.1`
mpmath	`1.3.0` (not used by these modules; used elsewhere in the repository)

SciPy is installed in the development environment but is **not** used by the desymmetrisation or dynamical modules; no result here depends on it.

7.3 Data and figures

The measured table, the coverage table and the false-positive rate are produced by `scripts/generate_desymmetrisation_figures.py` and pinned by `tests/test_desymmetrisation_figures.py`. Every number in the figures is computed at draw time; none is typed into the plotting code.

figure	file	contents
Fig. 1	`output/desymmetrisation.png`	$\langle r \rangle$ against unresolved sectors, 300 seeds per point
Fig. 2	`output/desymmetrisation_warning.png`	one dot per spectrum, four sector counts, both classes, with the Poisson window and the contamination floor
Fig. 3	`output/desymmetrisation_coverage.png`	measured coverage against the nominal 95% level
Fig. 4	`output/desymmetrisation_flow.png`	sector populations under the flow, commuting and cross-sector starts

8. Relation to prior work

reference	contribution	what this adds
Giraud et al. [1]	the phenomenon: superpositions bias spacing statistics	the measured table, the entry points, the posterior, the coverage
Rosenzweig–Porter [3]	block superpositions in random-matrix theory	the diagnostic form, not the theory
Mehta [2]	the reference distributions	the calibration values
Atas et al. [5]	the r-ratio as a statistic	the inversion attempt and its measured failure

The phenomenon is not new. The instrument is.

9. Conclusion

A spacing statistic measured on a Hamiltonian that commutes with an involution is measured on a superposition. The standard $\langle r \rangle$ cannot distinguish a Poisson spectrum from a union of unresolved sectors, and the failure is silent.

We provide a diagnostic that verifies the symmetries, splits the spectrum, flags contamination, and estimates the sector count as a posterior with a measured coverage. The single-sector case is decisive; the two-sector case is defensible as a set but not as a point; from three sectors upward the count is not recoverable. Two-sector GOE and GUE are indistinguishable. The false-positive rate is stated. The limits are reported rather than hidden.

The companion dynamical analysis shows that the double-bracket flow preserves sector populations when the state commutes with the involution, but selects a sector when it does not — so the flow cannot repair a contaminated spectrum, because the class is a property of the Hamiltonian and the Hamiltonian does not evolve.

The rule is simple: desymmetrize before testing the class.

Appendix A: The full measured table

At 300 seeds per point, `60` levels per sector, with the per-spectrum standard deviation of the mean $\langle r \rangle$:

sectors	GOE $\langle r \rangle$	GOE std	GUE $\langle r \rangle$	GUE std
1	`0.5309`	`0.0407`	`0.5999`	`0.0342`
2	`0.4260`	`0.0317`	`0.4209`	`0.0327`
3	`0.4036`	`0.0255`	`0.3989`	`0.0247`
4	`0.3960`	`0.0209`	`0.3933`	`0.0212`
6	`0.3907`	`0.0168`	`0.3889`	`0.0171`

The two-sector gap between classes (`0.0051`) is smaller than either standard deviation by a factor of six, which is the quantitative form of "the class is erased".

Appendix B: The coverage table

At a nominal 95% interval, 60 spectra per row:

true sectors	coverage	posterior on the truth
1	`1.00`	`0.98`
2	`0.93`	`0.65`
3	`0.87`	`0.24`
4	`0.98`	`0.17`
5	`1.00`	`0.21`
6	`0.97`	`0.20`
Poisson	`0.78`	`0.20`

The Poisson row is the false-positive rate of the posterior: it reports a finite count with probability `0.80`.

Appendix C: The false-positive rate

test	result
pure single-sector GOE spectra below the floor, `d = 60`	`4` of `40`
pure single-sector GUE spectra below the floor, `d = 60`	`0` of `40`
false-positive rate, GOE	`10%`

The floor is `0.48`, not the reference value `0.531`: the finite-size dependence of the one-sector value (`0.531` at `d = 60`, `0.519` at `d = 100`) is why the floor is set below the reference.

References

1. Giraud, O. et al. *Probing symmetries of quantum many-body systems through gap ratio statistics*. Phys. Rev. X **12**, 011006 (2022); arXiv:2008.11173.
2. Mehta, M. L. *Random Matrices*. 3rd ed., Elsevier (2004).
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5. Atas, Y. Y., Bogomolny, E., Giraud, O. and Roux, G. *Distribution of the ratio of consecutive level spacings in random matrix ensembles*. Phys. Rev. Lett. **110**, 084101 (2013).
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Disclaimer

This manuscript describes a diagnostic instrument, not a result about the Riemann hypothesis. The phenomenon it addresses — that unresolved symmetries bias spacing statistics toward Poisson — is standard quantum-chaos practice. What is local is the measured table, the entry points, the posterior's measured coverage, and the recorded limits.

The diagnostic detects contamination. It does not count sectors reliably above two, and it does not distinguish GOE from GUE at two sectors. Those limits are stated, not hidden.

The rule is: desymmetrize before testing the class.